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Course:

CMSC4

Section:

Term:

☐ Open Sections Only

Credit:

Level: ☒ All ☐ Undergraduate ☐ Graduate

Instructor(last,first):

Delivery: ☐ Face-to-Face ☐ Blended Learning ☐ Online

Start Time: :

Location/Program:

Course Days: ☐ Monday ☐ Tuesday ☐ Wednesday ☐ Thursday ☐ Friday

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Courses - Fall 2026



CMSC

Computer Science [Department Site](#)

CMSC411

(Perm Req)

Computer Systems Architecture

[Syllabus Repository \(0\)](#)



Credits: 3

Grad.Meth: Reg

Open Seats as of
06/08/2026 at 10:30 PM

QUESTIONS? ASK SHELLY

Prerequisite: Minimum grade of C- in CMSC330; or must be in the (Computer Science (Doctoral), Computer Science (Master's)) program.

Restriction: Permission of CMNS-Computer Science department.

Credit only granted for: ENEE446 or CMSC411.

Input/output processors and techniques. Intra-system communication, buses, caches. Addressing and memory hierarchies. Microprogramming, parallelism, and pipelining.

[Show Sections](#)

CMSC414

(Perm Req)

Computer and Network Security

[Syllabus Repository \(6\)](#)



Credits: 3

Grad Meth: Reg

Prerequisite: Minimum grade of C- in CMSC330 and CMSC351; or must be in the (Computer Science (Doctoral), Computer Science (Master's)) program.

Restriction: Permission of CMNS-Computer Science department.

Credit only granted for: CMSC414 or ENEE457.

An introduction to the topic of security in the context of computer systems and networks. Identify, analyze, and solve network-related security problems in computer systems. Fundamentals of number theory, authentication, and encryption technologies, as well as the practical problems that have to be solved in order to make those technologies workable in a networked environment, particularly in the wide-area Internet environment.

[Show Sections](#)

CMSC417

(Perm Req)

Computer Networks

[Syllabus Repository \(3\)](#)



Credits: 3

Grad Meth: Reg

Prerequisite: Minimum grade of C- in CMSC351 and CMSC330; and permission of CMNS-Computer Science department. Or must be in the (Computer Science (Doctoral), Computer Science (Master's)) program.

Computer networks and architectures. The OSI model including discussion and examples of various network layers. A general introduction to existing network protocols. Communication protocol specification, analysis, and testing.

[Show Sections](#)

CMSC420

(Perm Req)

Advanced Data Structures

[Syllabus Repository \(1\)](#)



Credits: 3

Grad Meth: Reg

Prerequisite: Minimum grade of C- in CMSC351 and CMSC330; and permission of CMNS-Computer Science department. Or must be in the (Computer Science (Doctoral), Computer Science (Master's)) program.

Description, properties, and storage allocation functions of data structures including balanced binary trees, B-Trees, hash tables, skiplists, tries, KD-Trees and Quadrees. Algorithms for manipulating structures. Applications from areas such as String Processing, Computer Graphics, Information Retrieval, Computer Networks, Computer Vision, and Operating Systems.

[Show Sections](#)

CMSC421

(Perm Req)

Introduction to Artificial Intelligence

[Syllabus Repository \(2\)](#)



Credits: 3

Grad Meth: Reg

Prerequisite: Minimum grade of C- in CMSC351 and CMSC330; and permission of CMNS-Computer Science department. Or must be in the (Computer Science (Doctoral), Computer Science (Master's)) program.

Introduces a range of ideas and methods in AI, varying semester to semester but chosen largely from: automated heuristic search, planning, games, knowledge representation, logical and statistical inference, learning, natural language processing, vision, robotics, cognitive modeling, and intelligent agents. Programming projects will help students obtain a hands-on feel for various topics.

[Show Sections](#)

CMSC422

(Perm Req)

Introduction to Machine Learning

[Syllabus Repository \(0\)](#)



Credits: 3

Grad Meth: Reg

Prerequisite: Minimum grade of C- in CMSC320, CMSC330, and CMSC351; and 1 course with a minimum grade of C- from (MATH240, MATH341, MATH461, ENEE290); and permission of CMNS-Computer Science department.

Machine Learning studies representations and algorithms that allow machines to improve their performance on a task from experience. This is a broad overview of existing methods for machine learning and an introduction to adaptive systems in general. Emphasis is given to practical aspects of machine learning and data mining.

[Show Sections](#)

CMSC423

(Perm Req)

Computational Genomics

[Syllabus Repository \(1\)](#)



Credits: 3

Grad Meth: Reg

Prerequisite: Minimum grade of C- in CMSC351 and CMSC330; and permission of CMNS-Computer Science department.

Computers have revolutionized modern biological research by enabling the management and analysis of the large amounts of data generated through high-throughput experiments. This course provides insight into some of the biological problems that can be addressed using computational methods and introduces the main data structures and algorithms used to solve problems related to biological sequences such as DNA, RNA, and

proteins. Many of the techniques taught in this course are applicable beyond the bioinformatics domain, as they are relevant to and commonly used in other areas of computer science. No knowledge of biology is required for this course.

[Show Sections](#)

CMSC424

(Perm Req)

Database Design

[Syllabus Repository_\(1\)](#)



Credits: 3

Grad Meth: Reg

Prerequisite: Minimum grade of C- in CMSC351 and CMSC330; and permission of CMNS-Computer Science department. Or must be in the (Computer Science (Doctoral), Computer Science (Master's)) program.

Students are introduced to database systems and motivates the database approach as a mechanism for modeling the real world. An in-depth coverage of the relational model, logical database design, query languages, and other database concepts including query optimization, concurrency control; transaction management, and log based crash recovery. Distributed and Web database architectures are also discussed.

[Show Sections](#)

CMSC425

(Perm Req)

Game Programming

[Syllabus Repository_\(0\)](#)



Credits: 3

Grad Meth: Reg

Prerequisite: Minimum grade of C- in CMSC330 and CMSC351.

Restriction: Permission of CMNS-Computer Science department.

An introduction to the principles and practice of computer game programming and design. This includes an introduction to game hardware and systems, the principles of game design, object and terrain modeling, game physics, artificial intelligence for games, networking for games, rendering and animation, and aural rendering. Course topics are reinforced through the design and implementation of a working computer game.

[Show Sections](#)

CMSC426

(Perm Req)

Computer Vision

[Syllabus Repository_\(0\)](#)



Credits: 3

Grad Meth: Reg

Prerequisite: Minimum grade of C- in CMSC330 and CMSC351 and 1 course with a minimum grade of C- from (MATH240, MATH341, MATH461, ENEE290); or must be in the (Computer Science (Doctoral), Computer Science (Master's)) program; or permission of the instructor.

Restriction: Permission of CMNS-Computer Science department.

An introduction to basic concepts and techniques in computervision. This includes low-level operations such as image filtering and edge detection, 3D reconstruction of scenes using stereo and structure from motion, and object detection, recognition and classification.

[Show Sections](#)

CMSC430

(Perm Req)

Introduction to Compilers

[Syllabus Repository_\(0\)](#)



Credits: 3

Grad.Meth: Reg

Prerequisite: Minimum grade of C- in CMSC330 and CMSC351; and permission of CMNS-Computer Science department. Or must be in the (Computer Science (Doctoral), Computer Science (Master's)) program.

Topics include lexical analysis, parsing, intermediate representations, program analysis, optimization, and code generation.

[Show Sections](#)

CMSC431

(Perm Req)

Privacy Engineering

[Syllabus Repository_\(0\)](#)



Credits: 3

Grad.Meth: Reg, P-F, Aud

Prerequisite: Minimum grade of C- in CMSC330 and CMSC351; and permission of CMNS-Computer Science Department.

Credit only granted for: CMSC431 or CMSC498G.

Formerly: CMSC498G.

Explores the intersection of technology and privacy, including technical mechanisms for data privacy, legal and regulatory privacy regimes, and mediation of interpersonal privacy through technology.

[Show Sections](#)

CMSC433

(Perm Req)

Programming Language Technologies and Paradigms

[Syllabus Repository_\(0\)](#)



Credits: 3

Grad.Meth: Reg

Prerequisite: Minimum grade of C- in CMSC330; or must be in the (Computer Science (Doctoral), Computer Science (Master's)) program.

Restriction: Permission of CMNS-Computer Science department.

Programming language technologies (e.g., object-oriented programming), their implementations and use in software design and implementation.

[Show Sections](#)

CMSC434

(Perm Req)

Introduction to Human-Computer Interaction

[Syllabus Repository_\(5\)](#)





Credits: 3

Grad Meth: Reg

Prerequisite: Minimum grade of C- in CMSC330 and CMSC351; and permission of CMNS-Computer Science department. Or must be in the (Computer Science (Doctoral), Computer Science (Master's)) program.

Assess usability by quantitative and qualitative methods. Conduct task analyses, usability tests, expert reviews, and continuing assessments of working products by interviews, surveys, and logging. Apply design processes and guidelines to develop professional quality user interfaces. Build low-fidelity paper mockups, and a high-fidelity prototype using contemporary tools such as graphic editors and a graphical programming environment (eg: Visual Basic, Java).

[Show Sections](#)

CMSC435

(Perm Req)

Software Engineering

[Syllabus Repository \(0\)](#)



Credits: 3

Grad Meth: Reg

Prerequisite: 1 course with a minimum grade of C- from (CMSC412, CMSC417, CMSC420, CMSC430, CMSC433, ENEE447); and permission of CMNS-Computer Science department.

State-of-the-art techniques in software design and development. Laboratory experience in applying the techniques covered. Structured design, structured programming, top-down design and development, segmentation and modularization techniques, iterative enhancement, design and code inspection techniques, correctness, and chief-programmer teams. The development of a large software project.

[Show Sections](#)

CMSC436

(Perm Req)

Programming Handheld Systems

[Syllabus Repository \(0\)](#)



Credits: 3

Grad Meth: Reg

Prerequisite: Minimum grade of C- in CMSC330 and CMSC351; or must be in the (Computer Science (Doctoral), Computer Science (Master's)) program.

Restriction: Permission of CMNS-Computer Science department.

Fundamental principles and concepts that underlie the programming of handheld systems, such as mobile phones, personal digital assistants, and tablet computers. Particular emphasis will be placed on concepts such as limited display size, power, memory and CPU speed; and new input modalities, where handheld systems differ substantially from non-handheld systems, and thus require special programming tools and approaches. Students will apply these concepts and principles in the context of an existing handset programming platform.

[Show Sections](#)

CMSC451

(Perm Req)

Design and Analysis of Computer Algorithms

[Syllabus Repository \(1\)](#)



Credits: 3

Grad Meth: Reg

Prerequisite: Minimum grade of C- in CMSC351; and permission of CMNS-Computer Science department. Or must be in the (Computer Science (Doctoral), Computer Science (Master's)) program.

Fundamental techniques for designing efficient computer algorithms, proving their correctness, and analyzing their complexity. General topics include graph algorithms, basic algorithm design paradigms (such as greedy algorithms, divide-and-conquer, and dynamic programming), network flows, NP-completeness, and other selected topics in algorithms.

[Show Sections](#)

CMSC456

(Perm Req)

Cryptography

[Syllabus Repository \(1\)](#)



Credits: 3

Grad Meth: Reg

Prerequisite: (CMSC106, CMSC131, or ENEE150; or equivalent programming experience); and (2 courses from (CMSC330, CMSC351, ENEE324, or ENEE382); or any one of these courses and a 400-level MATH course, or two 400-level MATH courses); and Permission of CMNS-Mathematics department or permission of instructor.

Cross-listed with: MATH456, ENEE456.

Credit only granted for: MATH456, CMSC456 or ENEE456.

The theory, application, and implementation of mathematical techniques used to secure modern communications. Topics include symmetric and public-key encryption, message integrity, hash functions, block-cipher design and analysis, number theory, and digital signatures.

[Show Sections](#)

CMSC460

(Perm Req)

Computational Methods

[Syllabus Repository \(2\)](#)



Credits: 3

Grad Meth: Reg

Prerequisite: 1 course with a minimum grade of C- from (MATH240, MATH341, MATH461, ENEE290, MATH243); and 1 course with a minimum grade of C- from (MATH241, MATH340); and 1 course with a minimum grade of C- from (CMSC106, CMSC131); and 1 course with a minimum grade of C- from (MATH246, MATH341, ENEE290, MATH243).

Cross-listed with: AMSC460.

Credit only granted for: AMSC460, AMSC466, CMSC460, or CMSC466.

Basic computational methods for interpolation, least squares, approximation, numerical quadrature, numerical solution of polynomial and transcendental equations, systems of linear equations and initial value problems for ordinary differential equations. Emphasis on methods and their computational properties rather than their analytic aspects. Intended primarily for students in the physical and engineering sciences.

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CMSC466

(Perm Req)

Introduction to Numerical Analysis I



[Syllabus Repository \(0\)](#)



Credits: 3

Grad.Meth: Reg

Prerequisite: 1 course with a minimum grade of C- from (CMSC106, CMSC131); and minimum grade of C- in MATH410.

Cross-listed with: AMSC466.

Credit only granted for: AMSC460, CMSC460, AMSC466, or CMSC466.

Floating point computations, direct methods for linear systems, interpolation, solution of nonlinear equations.

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CMSC471

(Perm Req)

Introduction to Data Visualization

[Syllabus Repository \(1\)](#)



Credits: 3

Grad.Meth: Reg

Prerequisite: Minimum grade of C- in CMSC330 and CMSC351; and permission of CMNS-Computer Science Department.

Restriction: Permission of the CMNS-Computer Science Department.

Credit only granted for: CMSC471 or CMSC498O.

Formerly: CMSC498O.

Datasets are becoming increasingly large and complex, requiring intuitive ways to explore and interpret them quickly and efficiently. In this case, a picture is worth a thousand words: visualizations enable us to transform data into images that are easier to understand and reason about, compared to raw numbers and raw text.

Visualizations are critical tools in externalizing and organizing our knowledge and insights, whether to explore collected datasets to improve our understanding of the physical world, to assess and debug analysis/experimental workflows, or to present new and interesting results to diverse audiences. In this course we will study techniques and algorithms for creating effective visualizations based on principles from graphic design, perceptual psychology, and cognitive science. Students will learn how to design and build interactive visualizations for the web, using the D3.js (Data-Driven Documents) framework.

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CMSC473

(Perm Req)

Capstone in Machine Learning

[Syllabus Repository \(0\)](#)



Credits: 3

Grad.Meth: Reg

Prerequisite: Minimum grade of C- or higher in CMSC421 or CMSC422.

Recommended: Background or exposure to machine learning topics is strongly encouraged.

Restriction: Permission of instructor and Permission of CMSC - Computer Science department.

Credit only granted for: CMSC498P or CMSC473.

Formerly: CMSC498P.

Additional information: Students will be paired with project advisors from the UMD faculty or alternatively, an industry advisor. Students are encouraged to plan for projects results that can be published at academic conferences or will impact academic research.

Semester-long project course in which each student will identify and carry out a project related to machine learning, with the goal of publishing a research paper or software tool.

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CMSC474

(Perm Req)

Introduction to Computational Game Theory

[Syllabus Repository_\(0\)](#)



Credits: 3

Grad Meth: Reg

Prerequisite: Minimum grade of C- in CMSC351 and CMSC330; and permission of CMNS-Computer Science department. Or must be in the (Computer Science (Doctoral), Computer Science (Master's)) program.

Credit only granted for: CMSC474, ECON414, GVPT390 or GVPT399A.

Game theory deals with interactions among agents (either human or computerized) whose objectives and preferences may differ from the objectives and preferences of the other agents. It will also provide a comprehensive introduction to game theory, concentrating on its computational aspects.

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CMSC475

(Perm Req)

Combinatorics and Graph Theory

[Syllabus Repository_\(1\)](#)



Credits: 3

Grad Meth: Reg

Prerequisite: 1 course with a minimum grade of C- from (MATH240, MATH341, MATH461, ENEE290, MATH243); and 1 course with a minimum grade of C- from (MATH241, MATH340); and 1 course with a minimum grade of C- from (MATH310, CMSC250).

Cross-listed with: MATH475.

Credit only granted for: MATH475 or CMSC475.

General enumeration methods, difference equations, generating functions. Elements of graph theory, matrix representations of graphs, applications of graph theory to transport networks, matching theory and graphical algorithms.

Credit only granted for MATH475 or CMSC475.

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CMSC498A

(Perm Req)

Selected Topics in Computer Science

[Syllabus Repository_\(0\)](#)



Credits: 1 - 3

Grad Meth: Reg

An individualized course designed to allow a student or students to pursue a selected topic not taught as a part of the regular course offerings under the supervision of a Computer Science faculty member. Credit according to work completed.

Contact department for information to register for this course.

CMSC498C



(Perm Req)

Selected Topics in Computer Science; Blockchains, Applied Cryptography, and Cryptocurrency

[Syllabus Repository \(0\)](#)



Credits: 3

Grad.Meth: Reg

Prerequisite: Minimum grade of C- in CMSC330 and CMSC351; and permission of CMNS-Computer Science Department.

Topics include blockchains ranging from fundamentals like consensus, to privacy-preserving payments, smart contracts, and decentralized finance(DeFi). The course will also cover recent developments in applied cryptography that are in increasing industrial usage such as zero-knowledge proofs and a small amount on multi-party computation and fully homomorphic encryption. It will look at what it takes to take these technologies from academic theory to real-world usage.

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CMSC498R

(Perm Req)

Selected Topics in Computer Science; AI and Public Policy

[Syllabus Repository \(0\)](#)



Credits: 3

Grad.Meth: Reg

Prerequisite: Minimum grade of C- in CMSC351, CMSC330 and CMSC320

Advances in Artificial Intelligence are presenting significant opportunities and potential challenges to society. This course will discuss societal issues raised by AI and potential public policy responses to them. We will first discuss some of the latest innovations in AI and how they are being integrated into work and society. We will then discuss a range of issues raised, including fairness and equity in AI systems, security issues raised by AI, privacy concerns and data ownership and energy and environmental issues raised by fueling AI. Students will be expected to read a wide range of both technical and policy material, and to discuss these readings in class.

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CMSC498Z

(Perm Req)

Selected Topics in Computer Science; Differentiable Programming for Agentic AI and Physical Intelligence

[Syllabus Repository \(0\)](#)



Credits: 3

Grad.Meth: Reg

Jointly offered with: CMSC838B

PREREQUISITES: Minimum grade of C- in CMSC421 or CMSC422; and permission of the CMNS-Computer Science department.

This course is an introduction to differentiable Programming, a new programming paradigm in which a numerical program can be differentiated through automatic differentiation, allowing gradient-based optimization of parameters in the program. It has broad applications in Computer Graphics, Computer Vision, Deep Learning, Quantum Computing, System Control, and many more. The course assumes a good working knowledge of linear

algebra and differentiation. The course experience includes hands-on projects with differentiable programming for agentic AI systems and physical intelligence.

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CMSC499A

(Perm Req)

Independent Undergraduate Research

[Syllabus Repository \(0\)](#)



Credits: 1 - 3

Grad Meth: Reg

Students are provided with an opportunity to participate in a computer science research project under the guidance of a faculty advisor. Format varies. Students and supervising faculty member will agree to a research plan which must be approved by the department. As part of each research plan, students should produce a final paper delineating their contribution to the field.

Contact department for information to register for this course.

Flagship Institution of the [University System of Maryland](#)

University of Maryland, College Park, MD 20742, USA / Phone: 301.405.1000



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